Lab Report: Training dynamics of deep models

1 Introduction

In this report we compare several models on standard benchmarks. Each experiment was repeated 5 times with different random seeds. We trained each model for 100 epochs with a batch size of 64.

2 Results

Figure 1 shows the validation accuracy during training.

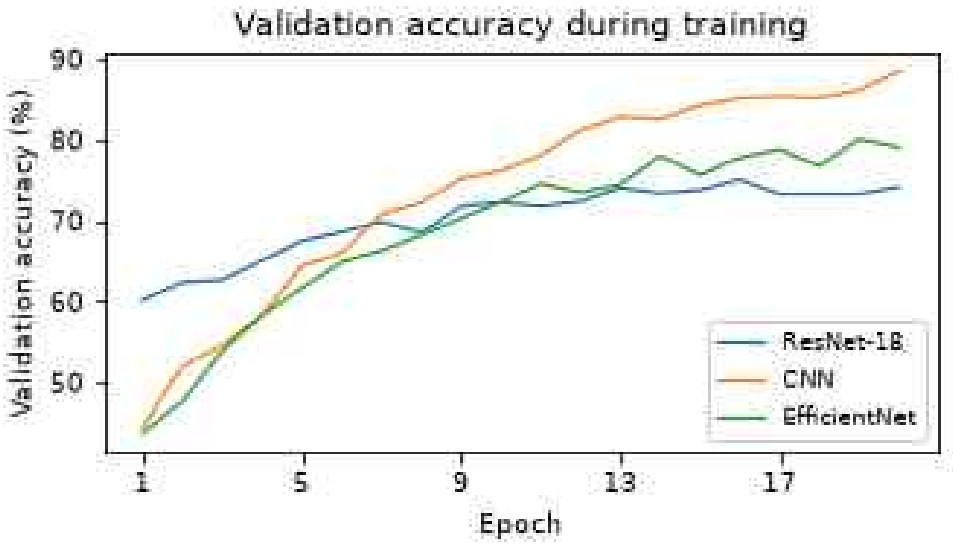


Figure 1: Validation accuracy during training for the models in this experiment.

At epoch 2, ResNet-18 leads CNN by 12.9 points (Figure 1). As shown in Figure 1, at epoch 4, EfficientNet achieves 82.9% validation accuracy. At epoch 20, EfficientNet achieves 80.1% validation accuracy.

Figure 2 shows the training loss.

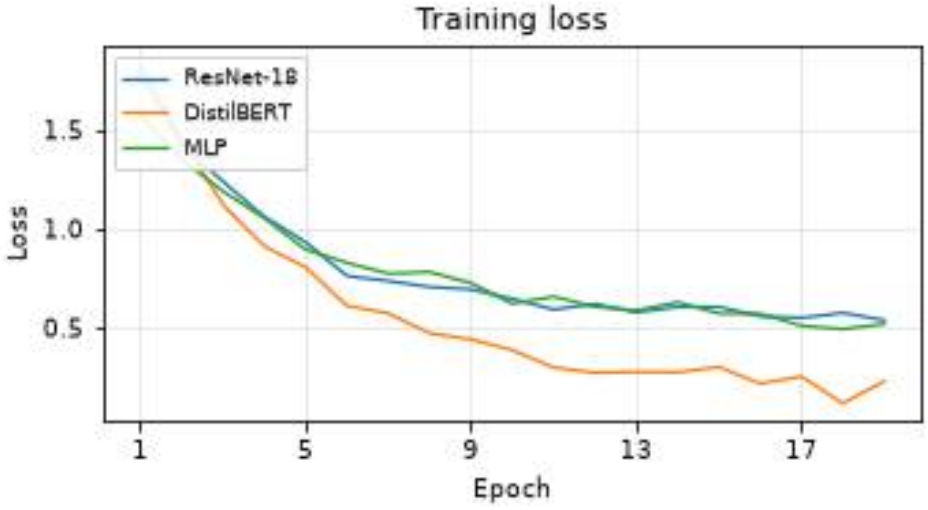


Figure 2: Training loss for the models in this experiment.

Epoch 3 sees ResNet-18 post a loss of 1.334. Epoch 18 sees ResNet-18 post a loss of 0.613. By epoch 9, the loss of ResNet-18 stands at 0.694 (Figure 2).

Figure 3 shows the test accuracy during training.

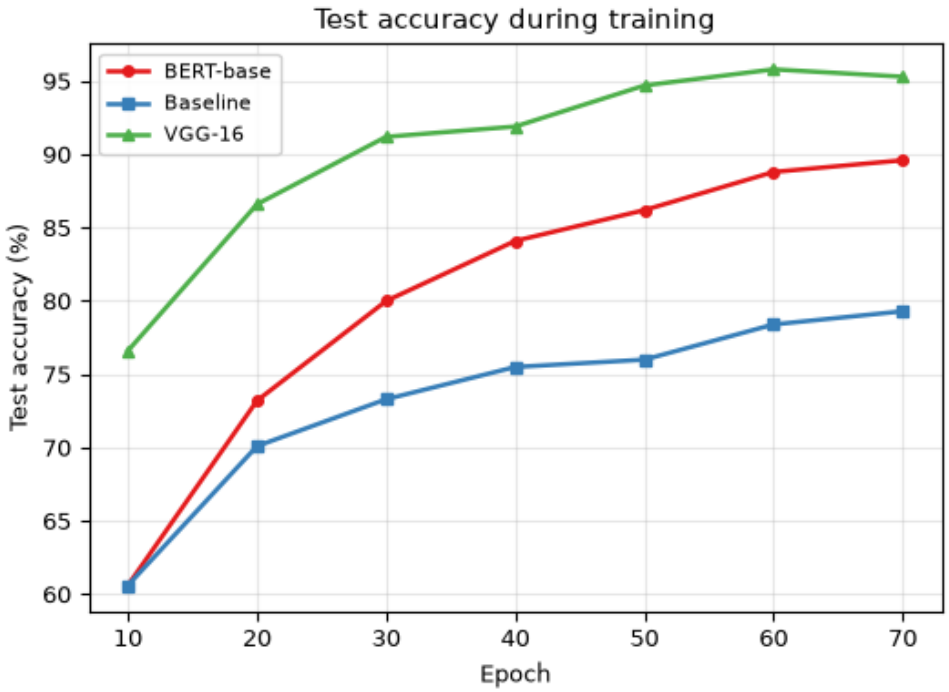


Figure 3: Test accuracy during training for the models in this experiment.

Baseline attains a test accuracy of 56.6% at epoch 50. Epoch 10 sees BERT-base post a test accuracy of 60.6% (Figure 3).

Figure 4 shows the test accuracy by dataset.

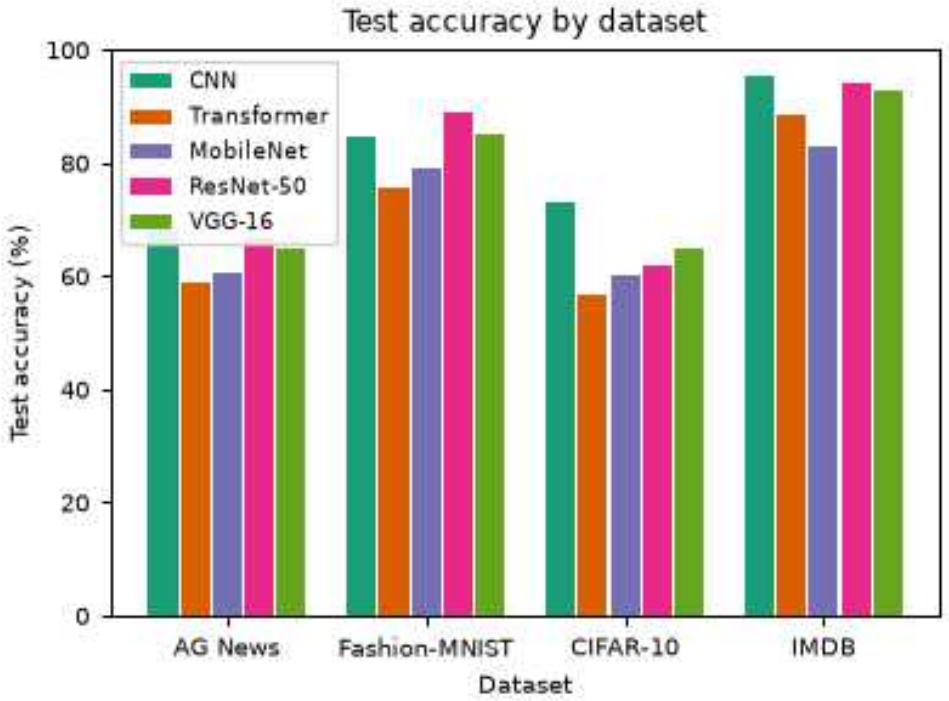


Figure 4: Test accuracy by dataset for the models in this experiment.

ResNet-50 beats VGG-16 by 1.4 points on IMDB. As shown in Figure 4, on Fashion-MNIST, VGG-16 leads Transformer by 10 points. MobileNet reaches 55.0% test accuracy on CIFAR-10.

3 Discussion

On STL-10, SVM obtains an accuracy of 88.9%. We note that the training was stable for all models.